MATH 114 - Computational Stochastics

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Contents

1 Overview

1.1 Introduction

Stochastic models describe real life phenomena like motion of the cell, shape of polymers stock prices, and more. What can we compute <u>analytically</u> and <u>numerically</u> from these models?

Key questions:

1. What is the distribution of the system?

2. How can we sample from the distribution?

3. How can we compute mean, variance, and higher moments of these systems?

Main topics:

1. Sampling RVs
2. Markov chain Monte Carlo (MCMC)
3. Stochastic differential equations (numerical & computation)

1.2 Sampling

- Generating "hard" RVs from "easy" RVs.
- Compute "hard" integrals.

$\boxed{ex}$ Area of the unit circle, $A = \pi$
Let $U_1 \sim U(-1,1)$ and $U_2 \sim U(-1,1)$
Let $x = \begin{cases} 0 \text{ if } (u_1, u_2) \text{ outside circle} \\ 1 \text{ if } (u_1, u_2) \text{ inside circle} \end{cases}$
$\implies P(X=1) = \frac{A}{4}, P(X=0) = 1 - \frac{A}{4}$
$\mathbb{E}(X) = \frac{A}{4}$
After N trials we generate $X_1, X_2, ..., X_n \quad \text{iid}$
By the strong law of large numbers

$$\lim_{N \to \inf} \frac{1}{N} \sum_{i=1}^{N} X_i = \frac{A}{4} = \frac{\pi}{4}$$

1.3 MCMC

Construct Markov chains to generate target distribution.
$\boxed{ex}$ Ising model

2 Markov Chains

2.1 Lecture 9

Def: A stochastic process is a family of RVs, $X_\epsilon \in S$, indexed by t, where $S \subseteq \mathbb{R}^d$ is a state space. Typically, $t \in [0, \infty)$, or $t\ in[a,b]$, or $t \in \{0,1,2,3,...\}$, etc.

- A discrete time stochastic process, is a sequence of RVs, $X_n \in S$, $n = 0, 1, 2, ...$ or $n = 1, 2, 3, ...$
- If S is finite or countably infinite then the stochastic process is a discrete state process.

First, we will examine discrete-time, discrete pace, stochastic process.
Two basic questions:

- Find the joint distribution

$$\begin{aligned}&P(X_0 = i_0, X_1 = i_1, ..., X_n = i_n)\\&i_0, i_1, ..., i_n \in S\end{aligned}$$

- Find the long-time behavior of the distribution of X_n:

$$P(x_n = i), i \in S, n >> 1$$

<u>Def:</u> Markov processes

- A stochastic process $\{X_n\}_{n=0}^{\infty}$ is a Markov process if

$$\begin{aligned}&P(X_{n+1} = i_{n+1}|x_0 = i_0, ..., x_n = i_n)\\&= P(X_{n+1}i_{n+1}|X_n = i_n)\\&\forall n \geq 0, \forall i_0, ..., i_{n+1} \in S\end{aligned}$$

- A Markov process is time homogeneous if:

$$\begin{aligned}&P(x_{n+1} = j|x_n = i)\\&= P(X_1 = j|X_0 = i)\\&\forall n \geq 0, \forall i, j \in S\end{aligned}$$

  Here we consider discrete time/state, time homogeneous Markov Chains (MC).

<u>Def:</u> Transition probability

$$\begin{aligned}P(i, j) &= P(X_1 = j|X_0 = i)\\&= P(i \to j) \quad \forall i, j \in S\end{aligned}$$

Def: Transition matrix

$$P = [p(i,j)] \text{ if } S = \{1,2,3,...\} \text{ or } \{0,1,2,...\}$$
Here i,j are states and also are row/column labels

For any $n \geq 1, i_0, ..., i_n \in S$, we have:

$$\begin{aligned}
&P(X_0 = i_0, ..., X_n = i_n) \\
&= P(X_0 = i_0)P(X_1 = i_1|X_0 = i_0)P(X_2 = i_2|X_1 = i_1, X_0 = i_0) \\
&... \ P(X_n = i_n|X_{n-1} = i_{n-1}, ..., X_0 = i_0) \qquad \text{[conditional probability]}
\end{aligned}$$

$\boxed{ex}$ Let $X_0, X_1, ..., X_n \in S$ iid. Then $\{X_n\}_{n=0}^{\infty}$ is a time homogeneous MC. Let $S = \{1,2,...,N\}$. Then $p(i,j) = \mathbb{P}(X_1 = j|X_0 = i) = \mathbb{P}(X_1 = j) = p_j$

$$\implies p = \begin{bmatrix} p_1 & \dots & p_n \\ \vdots & \ddots & \vdots \\ p_1 & \dots & p_n \end{bmatrix}, \sum_{i=1}^{n} p_i = 1$$

2.2 Lecture 10

Recap:

- Discrete time/state, time homogeneous
- MC: $\{X_n\}_{n=0}^{\infty} \in S$
- $\mathbb{P}(X_{n+1} = i_{n+1}|X_0 = i_0, ..., X_n = i_n) = \mathbb{P}(X_{n+1} = i_{n+1}|X_n = i_n)$
  $= p(i_n, i_{n+1})$
- Transition probability $\mathbb{P}(X_1 = j|X_0 = i)$
  $= p(i,j) = p_{ij} \quad \forall i,j \in S$
- Transition matrix $\mathbb{P}[p(i,j)]$

$\boxed{ex}$ Two-state MC, e.g. state of phone $X_n \in S = \{0,1\}$

$$\begin{cases} X_n = 0 & \text{phone free at time n} \\ X_n = 1 & \text{phone busy at time n} \end{cases}$$

$\mathbb{P}(\text{free} \to \text{busy in time step}) = p$
$\mathbb{P}(\text{busy} \to \text{free in time step}) = q$
$\mathbb{P}(\text{free} \to \text{free in time step}) = 1 - p$

$\mathbb{P}(\text{busy} \to \text{busy in time step}) = 1 - q$

Transition matrix:

$$P = \begin{bmatrix} 1-p & p \\ q & 1-q \end{bmatrix}$$

e.g. $p = \frac{1}{2}, q = \frac{1}{3}$

$$P = \begin{bmatrix} \frac{1}{2} & \frac{1}{2} \\ \frac{1}{3} & \frac{2}{3} \end{bmatrix}$$

Let $\mathbb{P}(X_0 = 0) = \frac{1}{5}, \mathbb{P}(X_0 = 1) = \frac{4}{5}$
Then $\mathbb{P}(X_0 = 0, X_1 = 1, X_2 = 0)$
$= \mathbb{P}(X_0 = 0) \cdot p(0,1) \cdot p(1,0)$
$= \frac{1}{5} \cdot \frac{1}{2} \cdot \frac{1}{3} = \frac{1}{30}$
And $\mathbb{P}(X_0 = 0, X_2 = 0) = \sum_{i=0}^{1} \mathbb{P}(X_0 = 0, X_1 = i, X_2 = 0)$

$\boxed{ex}$ $P_0 = I$, $p_0(i,j) = \delta_{ij} = \begin{cases} 1 & i = j \\ 0 & i \neq j \end{cases}$
$P_1 = P$, $p_1(i,j) = p(i,j)$

The distribution vector π_n for X_n is

$$\pi_n = (\pi_n(0), \pi_n(1), ...)$$
$$\text{where } \pi_n(i) = \mathbb{P}(X_n = i), i \in S$$
$$\text{Note that } \pi_n \text{ is a probability vector}$$

Property: Let $X_n \in S = \{1, 2, 3, ...\}, n \geq 0$ be the time homogeneous Markov Chain with transition matrix P. Then

- P is a stochastic matrix
- $P \cdot \mathbb{I} = \mathbb{I}$ where $\mathbb{I} = \begin{bmatrix} 1 \\ 1 \\ \vdots \\ 1 \end{bmatrix}$
- 1 is an eigenvalue of P with eigenvector $\mathbb{I}$

- $P_{m+n} = P_m \cdot P_n$ (Chapman-Kolmogorov equation)
- $P_n = P^n$
- $\pi_n = \pi_0 P_n = \pi_0 P^n$
- $\pi_{n+1} = \pi_n P,\ \forall n \geq 0$

Proof: direct verification
e.g. $\mathbb{P}(X_1 = j) = \sum_{i \in S} \mathbb{P}(X_0 = i)\mathbb{P}(X_1 = j | X_0 = i)$
$\pi_1(j) = \sum \pi_0(j) P_{ij}$
Long-run behavior, invariant distribution

Def: Let $P = [p(i,j)]$ be the transition matrix for Markov Chain $\{X_n\}_{n=0}^{\infty}$ with $S = \{1, 2, ..., N\}$. A probability vector $\pi \in \mathbb{R}^n$ is a stationary or invariant distribution if $\pi^p = \pi$

2.3 Lecture 11

- Recap: Discrete time/state, time homogeneous Markov Chains. Transition matrix $P = [p(i,j)],\ p(i,j) = \mathbb{P}(X_1 = j | X_j = i),\ p_n(i,j) = \mathbb{P}(X_n = j | X_0 = i)$
- Markov Chains $\to$ MCMC

$\boxed{ex}$ 3 state MC

$$P = \begin{bmatrix} \frac{3}{4} & \frac{1}{4} & 0 \\ \frac{1}{8} & \frac{2}{3} & \frac{5}{24} \\ 0 & \frac{5}{6} & \frac{1}{6} \end{bmatrix}$$
$$S = \{1, 2, 3\}$$
$$\mathbb{P}(X_1 = 1 | X_0 = 1) = \frac{3}{4}, \text{etc.}$$
$$\lim_{n \to \infty} P^n = \begin{bmatrix} \pi \\ \pi \\ \pi \end{bmatrix},\ \pi = (\frac{2}{11}, \frac{4}{11}, \frac{5}{11})$$
$$\text{Set } \pi_0 = (\mathbb{P}(X_0 = 1), \mathbb{P}(X_0 = 2), \mathbb{P}(X_0 = 3))$$
$$= \mathbb{P}(X_0 = i)$$

Then

$$\begin{aligned}
\pi_n &= \pi_0 p^n \\
\lim_{n\to\infty} \pi_n &= \pi_0 \lim_{n\to\infty} p^n \\
&= \pi_0 \begin{bmatrix} \pi \\ \pi \\ \pi \end{bmatrix} \\
&= \left(\pi(1)\cdot \sum_i \pi_0(i),\ \pi(2)\cdot \sum_i \pi_0(i), ...\right) \\
&= \pi
\end{aligned}$$

Remarks

- $\lim_{n\to\infty} \mathbb{P}(X_n = j) - \pi(j) \forall j$
-

$$\begin{aligned}
\pi_n &= \pi_{n-1} P \\
\lim_{n\to\infty} \pi_n &= \lim_{n\to\infty} \pi_{n-1} P \\
\pi &= \pi P
\end{aligned}$$

- π is a left eigenvector of P with eigenvalue 1.

Def: Let $P = [p(i,j)]$ be the transition matrix for MC $\{X_n\}_{n=0}^{\infty}$ with $S = \{1, ..., N\}$. A probability vector $\pi \in \mathbb{R}^n$ is a stationary or invariant distribution for X_n (or P) if: $\pi = \pi P$

Remarks
Above example shows $\lim_{n\to\infty} \pi_n =$ invariant distribution (with some conditions).

Questions: invariant distribution.

- When do they exist?
- Are they unique?
- Roughly: typically have unique invariant distribution unless reducible or periodic.

Def: A finite state time-homogeneous MC $X_n(n \geq 0$ with transition matrix P is regular if there exists an integer $k \geq 1$, such that $P^k > 0$, i.e. all entries of P^k are strictly > 0.

Thm
Let $X_n(n \geq 0)$ be a regular N-state Markov Chain with transition matrix P. Then:

1. $\{X_n\}_{n=0}^{\infty}$ has a unique stationary distribution $\pi,\ \pi > 0$
2. $\lim_{n\to\infty} P^n = \begin{bmatrix} \pi \\ \vdots \\ \pi \end{bmatrix}$, i.e. $\lim_{n\to\infty} p_n(i,j) = \pi(j)$
3. For any initial distribution $\pi_0, \lim_{n\to\infty} \pi_0 P^n = \pi$ i.e. $\lim_{n\to\infty} \mathbb{P}(X_n = j) = \pi(j)\ \forall j$

Remarks

- 2. $\implies \lim\limits_{n\to\infty} \mathbb{P}(X_n = j | X_0 = i) = \pi(j)$ independent of i
- From 2.:

$$\lim_{n\to\infty} \frac{1}{n} \sum_{k=0}^{n-1} p_k(i,j) = \lim_{n\to\infty} p_n(i,j) = \pi(j)$$

$$\begin{aligned}
\text{And } \frac{1}{n} \sum_{k=0}^{n-1} p_k(i,j) &= \frac{1}{n} \sum_{k=0}^{n-1} \mathbb{P}(X_k = j | X_0 = i) \\
&= \frac{1}{n} \sum_{k=0}^{n-1} \mathbb{E}[\mathbb{I}[X_k = j] | X_0 = i] \\
&= \mathbb{E}\Big[\frac{1}{n} \sum_{k=0}^{n-1} \mathbb{I}(X_k = j) | X_0 = i\Big] \\
&\to \pi(j) \text{ as } n \to \infty
\end{aligned}$$

Which means the mean fraction of time spent in $J \to \pi(j)$

How do we find π?

- Solve $\pi P = \pi$
- Diagonalize $P = U\Lambda U^{-1}$

$$\begin{aligned}\Rightarrow \lim_{n\to\infty} P^n &= \lim_{n\to\infty}(U\Lambda U^{-1})^n \\ &= U(\lim_{n\to\infty}\Lambda^n)U^{-1} \\ &= \begin{bmatrix}\pi \\ \vdots \\ \pi\end{bmatrix}\end{aligned}$$

2.4 Lecture 12

Recap:

- Discrete time/space MC $\{X_n\}_{n=0}^{\infty}, X_n \in S$
- $P = [p(i,j)], p(i,j) = \mathbb{P}(X_1 = j | X_0 = i)$
- $\pi_0 = \mathbb{P}(X_0 = i),\ \pi_n = \mathbb{P}(X_n = i)\quad$ (pmf X_n)
- Stationary distribution π s.t. $\pi P = \pi$
- Regular: $\exists k \geq 1$ s.t. $P^k > 0 \Rightarrow$ unique stationary distribution $\pi > 0$
  And $\lim_{n\to\infty} P^n = \begin{bmatrix}\pi \\ \vdots \\ \pi\end{bmatrix}$, $\lim_{n\to\infty}\pi_n = \pi$
- Detailed balance: probability vector λ s.t. $\lambda_i p(i,j) = \lambda_j p(j,i) \forall i,j \in S$
  $\Rightarrow \lambda P = \lambda \Rightarrow \lambda$ stationary with respect to P.

Remarks

1. Can view MC as evolution of probability vectors (pmfs): $\pi_0, \pi_1, ...$ with $\pi_{n+1} = \pi_n P$. This is hard to compute for $|S|$ large.
2. MC is a random sequence of states $X_0, X_1, ... \in S$ which is often easier to compute.

Recall: Discrete Inversion (Lecture 6)
e.g.: Generate discrete RV X

$$P(X = i) = p_i,\ \ i = 1, 2, 3$$
$$p_1 + p_2 + p_3 = 1$$

1. $U \sim U[0, 1]$
2. $k = min\{m | \sum_{i=1}^{m} p_i \geq U\}$
3. Set $X \leftarrow k$

<u>Algorithm:</u> Generate / simulate MC
Input: transition matrix P, initial distribution π_0
Output: MC $X_0, X_1, ... \in S, X_n \sim \pi_k = \pi_0 P^k$

1. Generate $X_0 \sim \pi_0$
2. For $k = 1, ..., n$, generate X_{n+1} from $p(X_k)$. i.e. pmf over S, given by the X_k'th row of P.
3. Repeat n times to get samples from π_k

<u>MCMC:</u> (R&K ch.6)

- Key idea: generate distribution π by simulating MC with stationary distribution π
- Often π is known up to a normalizing constant.
- $\lim_{n\to\infty} \frac{1}{n} \sum_{k=1}^{n} \mathbb{I}\{X_k = i\} = \pi(i)\ \forall i \in S$
- $\lim_{n\to\infty} \frac{1}{n} \sum_{k=1}^{n} g(X_k) = \mathbb{E}_\pi(g(x))$

We assume that

- RV $X \in S$ is finite, e.g. $S = \{1, 2, ..., N\}$
- Target pmf of X is $\pi = (\pi(1), ..., \pi_N), \pi(i) = \mathbb{P}(X = i)\ \forall i \in S$
- π has special form $\pi(i) = \frac{1}{z} b(i) > 0\ \forall i \in S$.
  Where $b(i) > 0$ are known and $z > 0$ is unknown norm.

$$\sum_{i\in S} \pi(i) = 1 \iff \sum_{i\in S} b(i) = z$$

Remarks

- z is hard to compute, (sum of $N = |S|$ terms)
- z is called a partition function

Algorithm: Metropolis

- First construct proposal transition matrix $Q = [q(i,j)]$, where Q is symmetric.
- Choose initial state $X_0 \in S$ (e.g. randomly).

Given $X_k = i \in S$

1. Sample $Y \in S$ from $q(i, \cdot)$, i.e. $\mathbb{P}(Y = y | X_k = i) = q(i, y)\ \forall y \in S$
   Assume $Y = j$ is observed.
2. Let $\alpha = \alpha_{ij} = \min\left(1, \frac{\pi(j)}{\pi(i)}\right) \in [0,1]$ and generate $U \sim U[0,1]$.
   If $U \leq \alpha$, accept Y, $X_{k+1} \leftarrow Y$
   If $U > \alpha$, reject Y, $X_{k+1} \leftarrow X_k$
   (Accept Y with probability α)
   Output: $X_0, X_1, X_2, ...$

Remarks

- $\{X_n\}_{n=0}^{\infty}$ is MC with $\lim_{n\to\infty} X_n \sim \pi$
- Can choose $Q = [q(i,j)]$ to be simple, e.g. uniform jumps, $q(i,j) = \frac{1}{N}$
- Only need the ratios $\frac{\pi(j)}{\pi(i)}$, so we don't need z.
- Call Y a proposed state.
- Call α the acceptance probability.
- In practice, $\alpha = a_{ij}$ is easy to compute.
- Can be extended to non-symmetric Q

$\boxed{ex}$ Id Ising model

$$\begin{aligned} S &= \{\sigma = (\sigma_1, ..., \sigma_m) | \sigma_i = -1 \text{ or } +1,\ 1 \leq i \leq m\} \\ &= \{-1, +1\}^m \\ |S| &= 2^m \text{ states} \end{aligned}$$